DISCOVER YOUR NEXT EVENT DESTINATION

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Abstract

Discover your next event destination is a responsive web-based event discovery platform designed to help users explore and filter local events efficiently. The application allows users to search events by city, category, price range, and sorting preferences. It incorporates modern web technologies including HTML semantic elements, CSS features such as Grid, Flexbox, animations, media queries, and JavaScript ES6 concepts like arrow functions, promises, async/await, and DOM manipulation. The system also includes interactive components such as modal popups, theme toggling (dark/light mode), toast notifications, and local storage integration for saving user preferences and bookmarked events. This is a simple web app which is user friendly.

Existing System

- Event information scattered across different platforms
- Difficult to find events quickly
- Limited filtering and search options
- No centralized event discovery platform
- Users cannot easily save favourite events
- Users must manually search multiple platforms.

Proposed System

- Centralized event discovery platform
- Search events by city
- Filter events by categories
- Sort events by date, price or name
- Save favourite events
- User friendly responsive interface
- Direct ticket booking feature.

Project details

Category:- Web Application

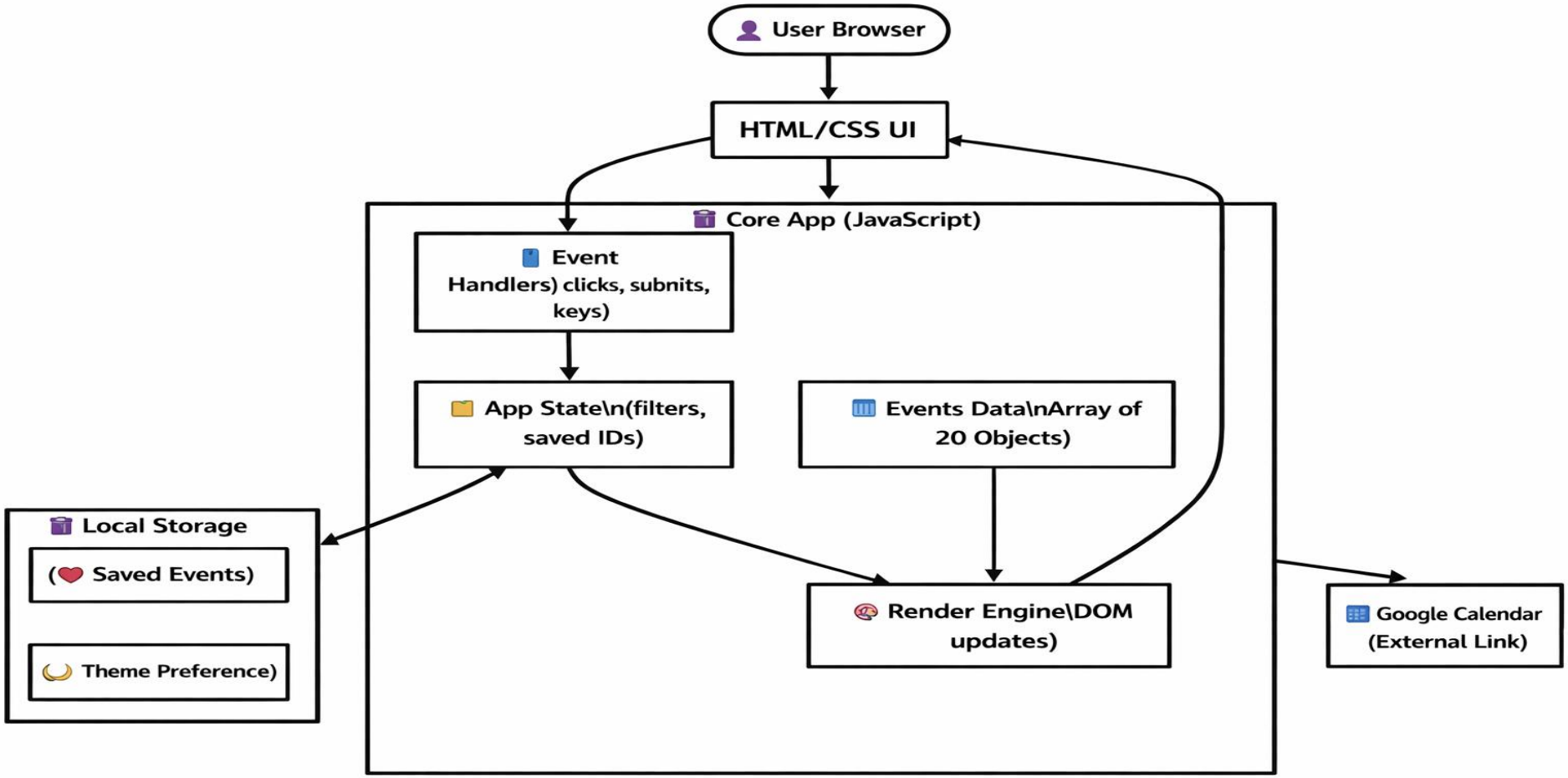
Purpose:- This code is designed to develop an interactive and user-friendly event discovery web application.
It allows users to search, filter, and sort events based on city, category, and price.
The application demonstrates modern web development concepts using HTML, CSS, and JavaScript. The project aims to provide a responsive and engaging platform for discovering local events efficiently.

Users:- Travelers, General users.

System Architecture

- **User Interface Layer**
- Built using **HTML, CSS, and JavaScript**
- Provides event search form, category filters, and event cards
- **Application Logic Layer**
- JavaScript handles **event filtering, sorting, and searching**
- **Data Layer**
- Event information stored as **JavaScript objects in arrays**
- **DSA Layer**
- Java program implements **searching and sorting algorithms**
- Uses **linked lists, stacks, queues, heaps, and hash tables** for data management
- **Output Layer**
- Displays filtered events and detailed information dynamically
- Provides interactive feedback through UI updates and notifications

System Architecture Diagram



Project Modules

- **Event Search Module**
- Allows users to search events using city and price filters
- **Category Filter Module**
- Filters events by categories such as music, gaming, sports, education, and arts
- **Event Display Module**
- Displays event name, location, date, and price
- **Event Details Module**
- Includes organizer, capacity, description, and date
- **Event Management Module**
- Implements searching, sorting, linked lists, stacks, queues, and hash tables

Objectives

- Develop an interactive event discovery website
- Provide easy access to event information
- Implement search and filtering features
- Improve user experience with responsive UI
- Demonstrate frontend development concepts

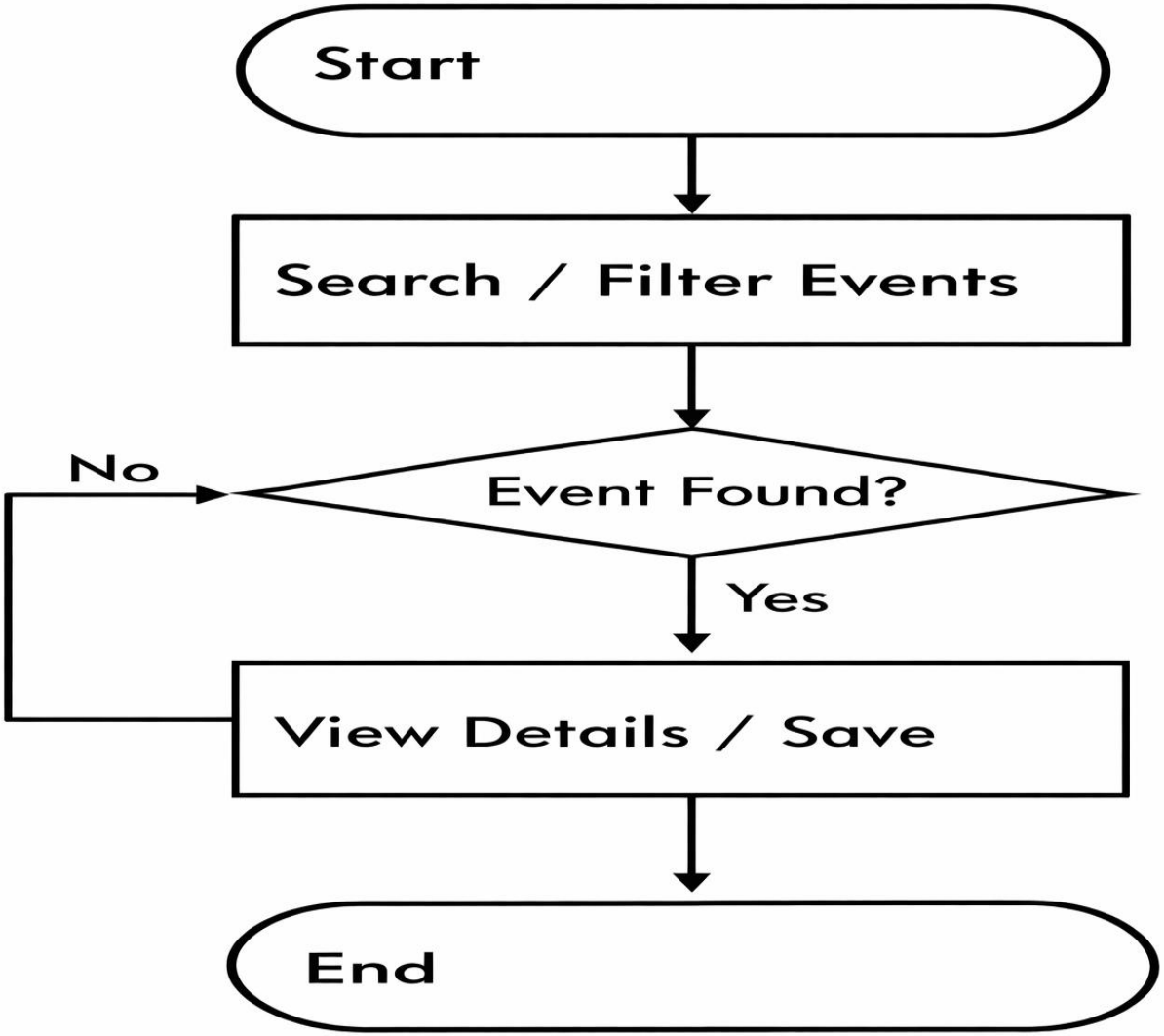
CO’s Attainments – FWD

- **CO1 – HTML Structure**
- Used HTML5 semantic elements to design the webpage structure
- **CO2 – CSS Styling**
- Applied CSS Grid, Flexbox, and responsive design
- Used animations, transitions, and media queries
- **CO3 – JavaScript Programming**
- Implemented variables, arrays, objects, and functions
- Stored event data using JavaScript objects
- **CO4 – DOM Manipulation**
- Used DOM methods and event listeners
- Dynamically updated event cards and modal windows
- **CO5 – Advanced Web Features**
- Implemented local Storage for saved events

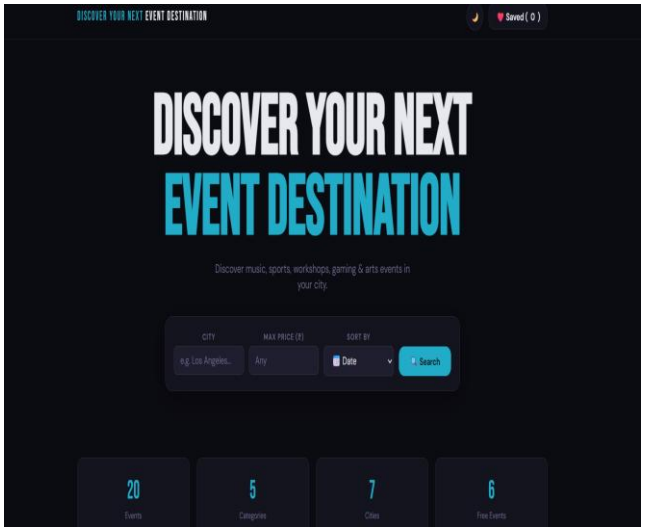
CO's Attainments – DSA

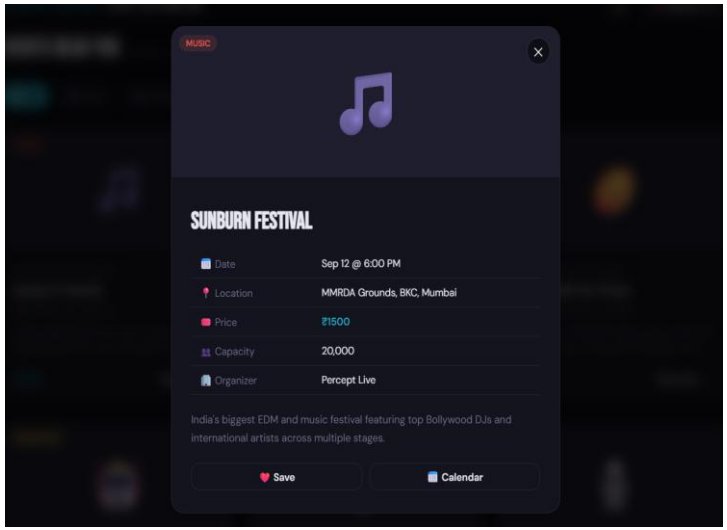
- **CO1 – Algorithm Analysis**
- Implemented searching and sorting algorithms
- Used Linear Search, Binary Search, Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, and Quick Sort
- Analyzed algorithm efficiency using Big-O notation
- **CO2 – Abstract Data Types**
- Implemented Singly Linked List, Doubly Linked List, and Circular Linked List
- **CO3 – Linear Data Structures**
- Implemented Stack and Queue structures
- Used **Circular Queue and Deque** for event processing
- **CO4 – Hash-Based Structures**
- Implemented Hash Tables using Chaining and Open Addressing

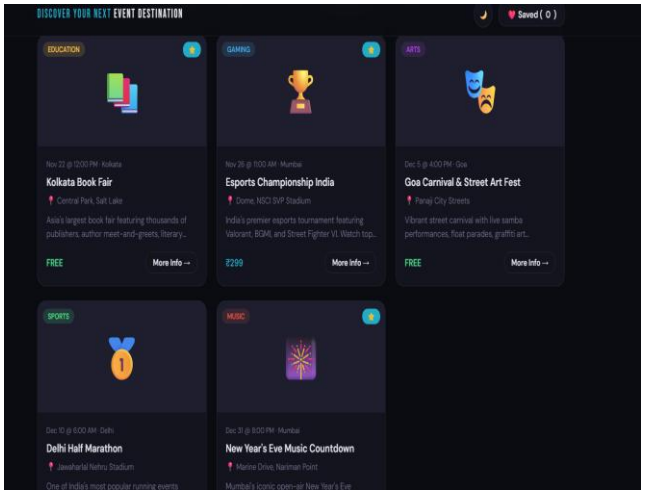
System Flowchart

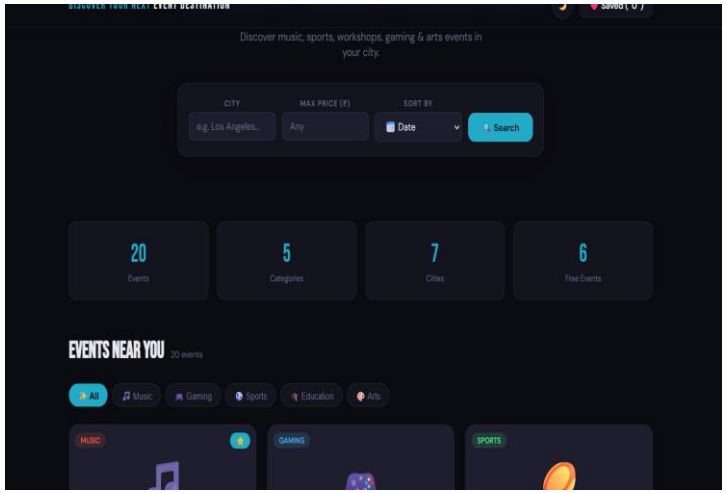


Screenshots – FWD









Screenshots – DSA

```
        boolean featured;

        Event(int id, String name, String category, String city,
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
? Circular Queue ? Registration Queue:
  Queue size: 5
  Processing: Alice, Bob

? Deque ? Recent & Upcoming Events:
  Deque contents:
    [ 2] India Gaming Expo (IGX)              | gaming    | Mumbai    | 2025-09-20 | ?499
    [ 1] Sunburn Festival                     | music     | Mumbai    | 2025-09-12 | ?1500
    [19] Delhi Half Marathon                  | sports    | Delhi     | 2025-12-10 | ?150
    [20] New Year's Eve Music Countdown       | music     | Mumbai    | 2025-12-31 | FREE
```

```
    // ==========================================================
    // CO1 | BINARY SEARCH
    // Topic: Searching — Binary Search  O(log n)
    // Pre-condition: array must be sorted by id
    // ==========================================================
    static int binarySearch(Event[] arr, int targetId) {
        int lo = 0, hi = arr.length - 1;
        while (lo <= hi) {                          // O(log n)
            int mid = lo + (hi - lo) / 2;
            if      (arr[mid].id == targetId) return mid;
            else if (arr[mid].id <  targetId) lo = mid + 1;
            else                              hi = mid - 1;
        }
        return -1;
    }

    // ==========================================================
    // CO1 | BUBBLE SORT
    // Topic: Sorting — Bubble Sort  O(n²)
```

```
import java.util.*;
import java.util.stream.*;

public class LocalPulseDSA {

    // ==========================================================
    // CO2 | ADT DESIGN — Event Entity (core data model)
    // Topic: Abstract Data Type using encapsulation
    // ==========================================================
    static class Event {
        int    id;
        String name;
        String category;   // music, gaming, sports, education, arts
        String city;
        String date;       // "YYYY-MM-DD"
        int    price;      // 0 = FREE
        int    capacity;
        String organizer;
        boolean featured;

        Event(int id, String name, String category, String city,
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
== CO6: Full System Summary ==============================

Total events loaded   : 20
Free events           : 7
Featured events       : 8
Cities covered        : 8
Algorithms demonstrated: LinearSearch, BinarySearch, BubbleSort,
    SelectionSort, InsertionSort, MergeSort, QuickSort, HeapSort
Data Structures used  : Array, SLL, DLL, CLL, Stack, CircularQueue,
    Deque, MaxHeap, MinHeap, HashTable(Chaining), HashTable(OA),
```

```
    }

    /** Traverse  O(n) */
    void traverse() {
        Node cur = head;
        while (cur != null) {
            System.out.println("   " + cur.data);
            cur = cur.next;
        }
    }

    /** Reverse list in-place  O(n) */
    void reverse() {
        Node prev = null, cur = head, next;
        while (cur != null) {
            next     = cur.next;
            cur.next = prev;
            prev     = cur;
            cur      = next;
        }
        head = prev;
```

Conclusion

- The project “Discover Your Next Event Destination” provides a platform to explore local events easily.
- Built using HTML, CSS, and JavaScript to create a responsive and interactive user interface.
- Users can search, filter, and view event details dynamically.
- Data Structures and Algorithms in Java were implemented for efficient event management.
- The project demonstrates practical knowledge of Front-End Web Development and DSA concepts.

Geo-Tag pictures

THANK YOU